

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Evaluating an Algebraic Integral

Evaluate the Integral:

$$\int_{\sqrt{2}}^{\sqrt{3}} \frac{x^5}{\sqrt{x^4 - 1}} dx$$

Solution: We begin by substituting $u = x^2$, which gives $du = 2x dx$, or $x dx = \frac{du}{2}$. We must also update the bounds: When $x = \sqrt{2}$, $u = 2$. When $x = \sqrt{3}$, $u = 3$. Rewrite the numerator to isolate $x dx$:

$$\int_{\sqrt{2}}^{\sqrt{3}} \frac{x^4 \cdot x}{\sqrt{x^4 - 1}} dx = \int_2^3 \frac{u^2}{\sqrt{u^2 - 1}} \frac{du}{2} = \frac{1}{2} \int_2^3 \frac{u^2}{\sqrt{u^2 - 1}} du$$

Next, we apply trigonometric (or hyperbolic) substitution. Let $u = \cosh(t)$, which means $du = \sinh(t) dt$, and $\sqrt{u^2 - 1} = \sinh(t)$. The integral becomes:

$$\frac{1}{2} \int \frac{\cosh^2(t)}{\sinh(t)} \sinh(t) dt = \frac{1}{2} \int \cosh^2(t) dt$$

Using the half-angle identity $\cosh^2(t) = \frac{\cosh(2t) + 1}{2}$:

$$\frac{1}{4} \int (\cosh(2t) + 1) dt = \frac{1}{4} \left(\frac{\sinh(2t)}{2} + t \right) = \frac{1}{4} (\sinh(t) \cosh(t) + t)$$

Substitute back u :

$$\frac{1}{4} \left(u \sqrt{u^2 - 1} + \operatorname{arccosh}(u) \right) = \frac{1}{4} \left(u \sqrt{u^2 - 1} + \ln |u + \sqrt{u^2 - 1}| \right)$$

Now evaluate from $u = 2$ to $u = 3$: Upper bound ($u = 3$): $\frac{1}{4} (3\sqrt{8} + \ln(3 + \sqrt{8})) = \frac{3\sqrt{2}}{2} + \frac{1}{4} \ln(3 + 2\sqrt{2})$ Lower bound ($u = 2$): $\frac{1}{4} (2\sqrt{3} + \ln(2 + \sqrt{3})) = \frac{\sqrt{3}}{2} + \frac{1}{4} \ln(2 + \sqrt{3})$

Subtracting the lower bound from the upper bound:

$$I = \frac{3\sqrt{2} - \sqrt{3}}{2} + \frac{1}{4} \ln \left(\frac{3 + 2\sqrt{2}}{2 + \sqrt{3}} \right)$$

Notice that $3 + 2\sqrt{2} = (1 + \sqrt{2})^2$, so $\frac{1}{4} \ln(3 + 2\sqrt{2}) = \frac{1}{2} \ln(1 + \sqrt{2})$. Thus, the solution simplifies to:

$$I = \frac{3\sqrt{2} - \sqrt{3}}{2} + \frac{1}{2} \ln(1 + \sqrt{2}) - \frac{1}{4} \ln(2 + \sqrt{3})$$

1.2 Hard Integral

Evaluating an Integral with Symmetry

Evaluate the Integral:

$$I = \int_0^{\pi/2} \ln(\sin(x) + \cos(x)) dx$$

Solution: First, we rewrite the sum of sine and cosine using the harmonic addition identity:

$$\sin(x) + \cos(x) = \sqrt{2} \cos\left(x - \frac{\pi}{4}\right)$$

Substituting this back into the integral:

$$I = \int_0^{\pi/2} \ln\left(\sqrt{2} \cos\left(x - \frac{\pi}{4}\right)\right) dx = \int_0^{\pi/2} \ln(\sqrt{2}) dx + \int_0^{\pi/2} \ln\left(\cos\left(x - \frac{\pi}{4}\right)\right) dx$$

The first integral evaluates easily:

$$\int_0^{\pi/2} \frac{1}{2} \ln(2) dx = \frac{\pi}{4} \ln(2)$$

For the second integral, we substitute $u = x - \frac{\pi}{4}$. The limits change from $0 \rightarrow \pi/2$ to $-\pi/4 \rightarrow \pi/4$:

$$\int_{-\pi/4}^{\pi/4} \ln(\cos u) du$$

Because $\ln(\cos u)$ is an even function, we can rewrite this as:

$$2 \int_0^{\pi/4} \ln(\cos u) du$$

It is a known identity involving Catalan's constant (G) that $\int_0^{\pi/4} \ln(\cos u) du = \frac{G}{2} - \frac{\pi}{4} \ln(2)$. Substituting this into our equation gives:

$$2 \left(\frac{G}{2} - \frac{\pi}{4} \ln(2) \right) = G - \frac{\pi}{2} \ln(2)$$

Now add the constant term evaluated earlier:

$$I = \frac{\pi}{4} \ln(2) + G - \frac{\pi}{2} \ln(2)$$

$$I = G - \frac{\pi}{4} \ln(2)$$

Differentials

2.1 Medium Differential

Coupled Linear ODEs

A particle is moving in the xy plane following the equations:

$$\begin{cases} x''(t) + 4x(t) = 0 \\ y''(t) + 2y'(t) + 5y(t) = 0 \end{cases}$$

Given the initial conditions $x(0) = y(0) = 0$, $x'(0) = 4$, and $y'(0) = 6$, determine the time $t > 0$ when $y(t) = x(t)$ other than at the origin.

Solution: We solve each ordinary differential equation separately.

For $x(t)$: The characteristic equation is $r^2 + 4 = 0$, giving purely imaginary roots $r = \pm 2i$.

$$x(t) = A \cos(2t) + B \sin(2t)$$

$x(0) = 0 \implies A = 0$, so $x(t) = B \sin(2t)$. Take the derivative: $x'(t) = 2B \cos(2t)$. $x'(0) = 2B = 4 \implies B = 2$.

$$x(t) = 2 \sin(2t)$$

For $y(t)$: The characteristic equation is $r^2 + 2r + 5 = 0$. By the quadratic formula, $r = -1 \pm 2i$.

$$y(t) = e^{-t}(C \cos(2t) + D \sin(2t))$$

$y(0) = 0 \implies C = 0$, so $y(t) = D e^{-t} \sin(2t)$. Take the derivative using the product rule: $y'(t) = D(-e^{-t} \sin(2t) + 2e^{-t} \cos(2t))$. $y'(0) = 2D = 6 \implies D = 3$.

$$y(t) = 3e^{-t} \sin(2t)$$

We need to find $t > 0$ such that $x(t) = y(t)$, and the position is not at the origin (meaning $\sin(2t) \neq 0$).

$$2 \sin(2t) = 3e^{-t} \sin(2t)$$

Since we rule out the origin, we can safely divide by $\sin(2t)$:

$$2 = 3e^{-t} \implies e^{-t} = \frac{2}{3} \implies e^t = 1.5$$

$$t = \ln(1.5)$$

2.2 Hard Differential

Second-Order Non-Homogeneous ODE

Solve the differential equation:

$$2 \frac{d^2 y}{dx^2} - 2y = \frac{\sin(x)}{e^x}$$

Given the initial conditions $y(0) = 2.3$ and $y'(0) = 2.5$, find the exact value of $y(\pi - 3)$.

Solution: The ODE can be written as $2y'' - 2y = e^{-x} \sin(x)$. First, find the homogeneous solution for $2y'' - 2y = 0 \implies y'' - y = 0$. The roots of the characteristic equation $r^2 - 1 = 0$ are $r = \pm 1$.

$$y_h(x) = C_1 e^x + C_2 e^{-x}$$

For the particular solution y_p , we guess $y_p(x) = e^{-x}(A \sin x + B \cos x)$. Differentiate $y_p(x)$ carefully using the product rule:

$$y_p' = e^{-x}((-A - B) \sin x + (A - B) \cos x)$$

$$y_p'' = e^{-x}(2B \sin x - 2A \cos x)$$

Substitute y_p and y_p'' into the original ODE:

$$2(e^{-x}(2B \sin x - 2A \cos x)) - 2(e^{-x}(A \sin x + B \cos x)) = e^{-x} \sin(x)$$

Match the coefficients for $\sin x$ and $\cos x$:

$$(4B - 2A) \sin x - (4A + 2B) \cos x = \sin x$$

This produces a system of equations: $4B - 2A = 1$ $-4A - 2B = 0 \implies B = -2A$ Substitute B into the first equation: $4(-2A) - 2A = 1 \implies -10A = 1 \implies A = -0.1$. $B = -2(-0.1) = 0.2$. The general solution is:

$$y(x) = C_1 e^x + C_2 e^{-x} + e^{-x}(-0.1 \sin x + 0.2 \cos x)$$

Apply initial conditions to find C_1 and C_2 : $y(0) = C_1 + C_2 + 0.2 = 2.3 \implies C_1 + C_2 = 2.1$. Find $y'(x)$: $y'(x) = C_1 e^x - C_2 e^{-x} + e^{-x}(-0.1 \sin x - 0.3 \cos x)$ $y'(0) = C_1 - C_2 - 0.3 = 2.5 \implies C_1 - C_2 = 2.8$. Adding the two equations yields $2C_1 = 4.9 \implies C_1 = 2.45$. Subtracting the equations yields $2C_2 = -0.7 \implies C_2 = -0.35$. The specific solution is:

$$y(x) = 2.45e^x - 0.35e^{-x} + e^{-x}(0.2 \cos x - 0.1 \sin x)$$

Evaluate at $x = \pi - 3$:

$$y(\pi - 3) = 2.45e^{\pi-3} - 0.35e^{3-\pi} + e^{3-\pi}(0.2 \cos(\pi - 3) - 0.1 \sin(\pi - 3))$$

Since $\cos(\pi - 3) = -\cos(3)$ and $\sin(\pi - 3) = \sin(3)$:

$$y(\pi - 3) = 2.45e^{\pi-3} - e^{3-\pi}(0.35 + 0.2 \cos(3) + 0.1 \sin(3))$$

Derivatives

3.1 Medium Derivative

Related Rates of a Volume of Revolution

A bowl is formed by rotating the curve $y = \frac{e^{2x}}{2(e^x+1)}$ around the y -axis for $0 \leq x \leq 2$. The bowl is filled with water at a rate of $\pi \text{ cm}^3 \text{ s}^{-1}$. Find the rate of change of the water height when the height is $\frac{9}{8} \text{ cm}$.

Solution: The volume V of a solid of revolution around the y -axis filled to height y is given by $V = \int_0^y \pi x^2 dy$. Differentiating with respect to y (via the Fundamental Theorem of Calculus) gives:

$$\frac{dV}{dy} = \pi x^2$$

We are given that $\frac{dV}{dt} = \pi$. Apply the chain rule $\frac{dV}{dt} = \frac{dV}{dy} \frac{dy}{dt}$:

$$\pi = \pi x^2 \frac{dy}{dt} \implies \frac{dy}{dt} = \frac{1}{x^2}$$

We must find x when $y = \frac{9}{8}$:

$$\frac{e^{2x}}{2(e^x+1)} = \frac{9}{8}$$

Cross-multiply to get $8e^{2x} = 18e^x + 18$, simplifying to $4e^{2x} - 9e^x - 9 = 0$. Let $u = e^x$. This is a quadratic in u : $4u^2 - 9u - 9 = 0$. Factoring: $(4u+3)(u-3) = 0$. Since $u = e^x > 0$, the only physical root is $u = 3$.

$$e^x = 3 \implies x = \ln(3)$$

Substitute x back into our rate of change equation:

$$\frac{dy}{dt} = \frac{1}{(\ln 3)^2} \text{ cm s}^{-1}$$

3.2 Hard Derivative

Steepest Descent and Multivariable Gradient

Let $g(x, y) = x^y + y^x + \tan^{-1}\left(\frac{x}{y}\right)$. Find the rate of change of g at the point $(1, 2)$ in the direction of steepest descent.

Solution: The direction of steepest descent is given by the negative gradient vector $-\nabla g(x, y)$, and the corresponding rate of change is $-\|\nabla g(x, y)\|$. Let's compute the partial derivatives evaluated at $(1, 2)$.

For $g_x(x, y)$:

$$g_x(x, y) = yx^{y-1} + y^x \ln(y) + \frac{1}{1 + (x/y)^2} \cdot \left(\frac{1}{y}\right) = yx^{y-1} + y^x \ln(y) + \frac{y}{x^2 + y^2}$$

Evaluate at $(1, 2)$:

$$g_x(1, 2) = 2(1)^1 + 2^1 \ln(2) + \frac{2}{1^2 + 2^2} = 2 + 2 \ln(2) + \frac{2}{5} = 2.4 + 2 \ln(2)$$

For $g_y(x, y)$:

$$g_y(x, y) = x^y \ln(x) + xy^{x-1} + \frac{1}{1 + (x/y)^2} \cdot \left(-\frac{x}{y^2}\right) = x^y \ln(x) + xy^{x-1} - \frac{x}{x^2 + y^2}$$

Evaluate at $(1, 2)$:

$$g_y(1, 2) = 1^2 \ln(1) + 1(2)^0 - \frac{1}{1^2 + 2^2} = 0 + 1 - \frac{1}{5} = 0.8$$

The gradient at $(1, 2)$ is $\nabla g = (2.4 + 2 \ln(2))\hat{i} + 0.8\hat{j}$. The rate of steepest descent is $-\|\nabla g\|$:

$$-\|\nabla g\| = -\sqrt{(2.4 + 2 \ln(2))^2 + (0.8)^2}$$

Expand the terms:

$$(2.4 + 2 \ln 2)^2 = 5.76 + 9.6 \ln 2 + 4(\ln 2)^2$$

$$(0.8)^2 = 0.64$$

Summing them up yields:

$$-\sqrt{6.4 + 9.6 \ln(2) + 4(\ln(2))^2}$$

Physics & Engineering

4.1 Quantum

Energy of Emitted Photon

An electron in a hydrogen-like singly ionized helium atom (He^+) undergoes a quantum transition from the energy level $n = 4$ down to the $n = 2$ level. Calculate the energy of the emitted photon in electron-volts (eV). Take the ground state energy of a hydrogen atom to be -13.6 eV.

Solution: The energy levels of a hydrogen-like ion are given by the Bohr formula:

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

For a singly ionized helium atom (He^+), the atomic number is $Z = 2$. Thus, $E_n = -13.6 \frac{2^2}{n^2} = -\frac{54.4}{n^2}$ eV.

Calculate the energy of the initial state ($n = 4$):

$$E_4 = -\frac{54.4}{4^2} = -\frac{54.4}{16} = -3.4 \text{ eV}$$

Calculate the energy of the final state ($n = 2$):

$$E_2 = -\frac{54.4}{2^2} = -\frac{54.4}{4} = -13.6 \text{ eV}$$

The energy of the emitted photon is the difference between these two states:

$$\Delta E = E_4 - E_2 = -3.4 - (-13.6) = 10.2 \text{ eV}$$

4.2 Electric

Motional EMF and Power Dissipation

A conducting rod 0.850 m long moves at 18.0 m s^{-1} perpendicular to a magnetic field of 0.650 T. The circuit resistance is 3.20Ω . Calculate the electrical power dissipated in watts.

Solution: First, calculate the motional electromotive force ($\mathcal{E}$) induced across the ends of the conducting rod moving through a magnetic field:

$$\mathcal{E} = Blv$$

Where $B = 0.650 \text{ T}$, $l = 0.850 \text{ m}$, and $v = 18.0 \text{ m s}^{-1}$.

$$\mathcal{E} = 0.650 \times 0.850 \times 18.0 = 9.945 \text{ V}$$

The power dissipated as heat in the circuit's resistance R is given by Joule heating:

$$P = \frac{\mathcal{E}^2}{R}$$

Substitute the calculated EMF and the given resistance (3.20Ω):

$$P = \frac{(9.945)^2}{3.20} = \frac{98.903025}{3.20} \approx 30.907 \text{ W}$$

Rounding to three significant figures, the power dissipated is **30.9 W**.

4.3 Mechanics

Horizontal Range of a Projectile

A particle is projected from the ground with an initial velocity of $u = 20 \text{ m s}^{-1}$ at an angle of $\theta = 30^\circ$ with the horizontal. Find the horizontal range R of the projectile in meters. Take $g = 9.8 \text{ m s}^{-2}$.

Solution: The horizontal range R of a projectile launched and landing on flat ground is given by the standard kinematic formula:

$$R = \frac{u^2 \sin(2\theta)}{g}$$

Substitute the given values $u = 20 \text{ m s}^{-1}$, $\theta = 30^\circ$, and $g = 9.8 \text{ m s}^{-2}$:

$$R = \frac{20^2 \sin(2 \cdot 30^\circ)}{9.8} = \frac{400 \sin(60^\circ)}{9.8}$$

Since $\sin(60^\circ) = \frac{\sqrt{3}}{2}$:

$$R = \frac{400 \left(\frac{\sqrt{3}}{2} \right)}{9.8} = \frac{200\sqrt{3}}{9.8}$$

Simplify the fraction:

$$R = \frac{2000\sqrt{3}}{98} = \frac{1000\sqrt{3}}{49} \text{ meters}$$

This yields an exact value. A numerical approximation would be roughly 35.35 m.